

CANDIDATE NAME _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2015

H2 BIOLOGY
9648

Paper 1

**Thursday
27 Aug 2015**

1 hour 15 minutes

Additional materials:
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Write your name and CG in the spaces at the top of this page.

On the Optimal Mark Sheet, enter your name, subject title, test name, class. For your index number, enter your full NRIC number. Shade the corresponding lozenges on the OMS according to the instructions given by the invigilators.

AT THE END OF THE EXAMINATION, HAND IN BOTH THE OMS AND QUESTION PAPER.

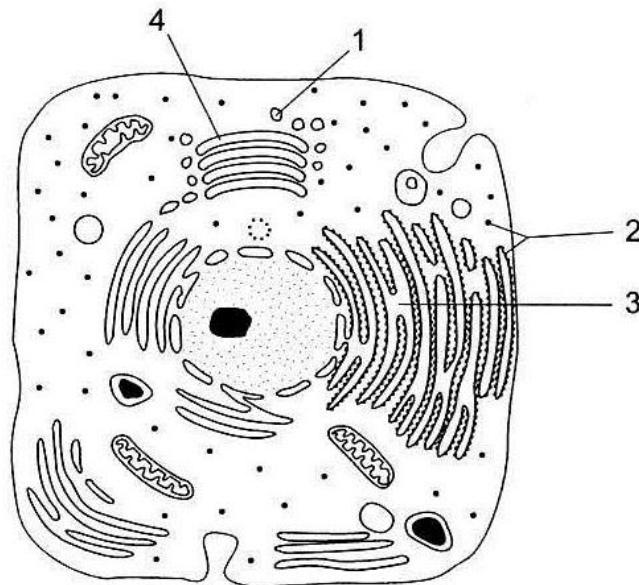
INFORMATION FOR CANDIDATES

There are **Forty (40) questions** in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C, D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done on the question paper.

This question paper consists of 24 printed pages.

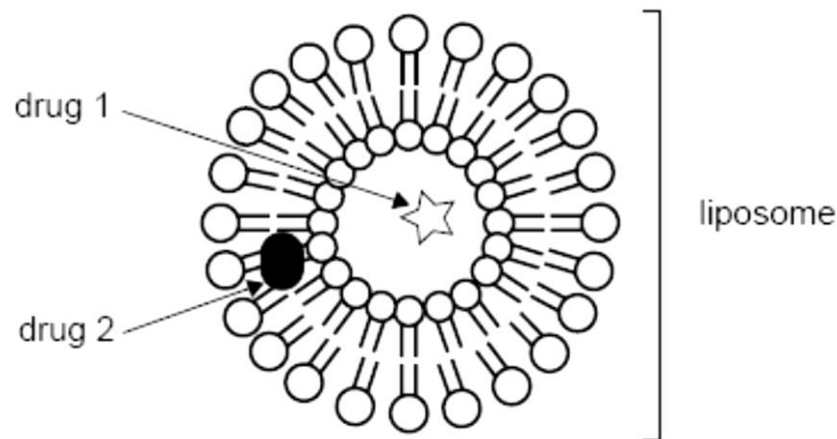
1) The figure below shows the structure of an animal cell.



Which of the following correctly identifies the functions of the labeled structures?

	Synthesizing polypeptides from amino acids	Transporting proteins	Carrying out glycosylation	Secreting digestive enzymes
A	2	1	4	3
B	1	4	2	3
C	2	3	4	1
D	1	3	2	4

- 2) Liposomes are spherical vesicles consisting of a single phospholipid bilayer. They can be designed to carry pharmaceutical drugs into cells. The figure below shows the structure of a liposome and two potential drugs, drug 1 and drug 2, that it can carry.



Which of the following statements are true?

- I. Liposomes can carry both hydrophobic and hydrophilic drugs.
- II. The interior of the liposome is aqueous.
- III. Liposomes are able to fuse with the cell surface membrane of a cell.
- IV. Liposome-based membranes have a rigid structure.

- A I and II only
- B II and III only
- C I, II and III only
- D II, III and IV only

- 3) Egg whites are rich in the globular protein albumin. When egg whites are cooked, they turn opaque and solid.

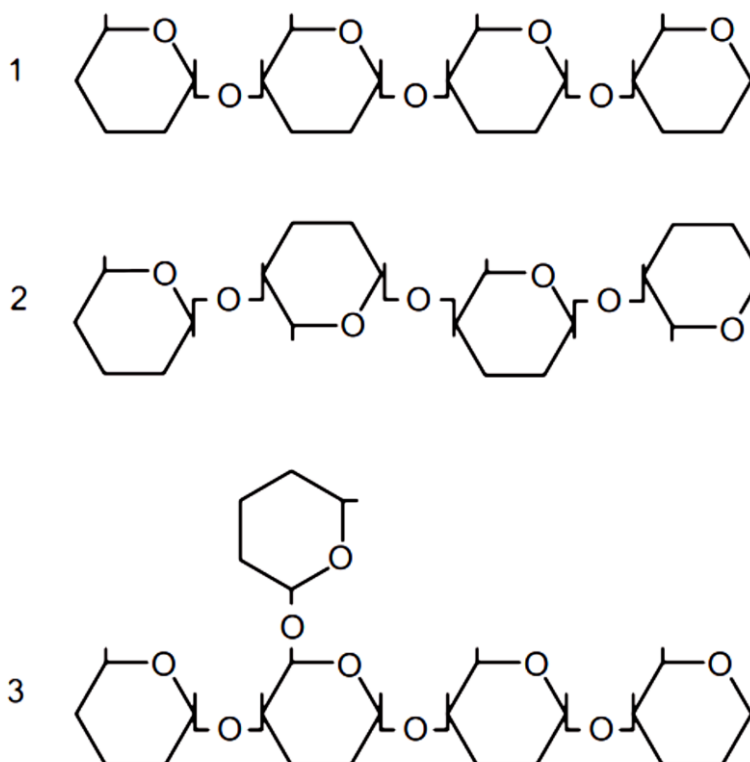
Which of the following statements provide an explanation for the observation to egg whites when cooked?

- A The disruption of hydrogen and ionic bonds between R groups of amino acid residues.
- B The disruption of peptide bonds between R groups of amino acid residues.
- C The disruption of hydrogen bonds between amino group and carboxyl groups of amino acid residues.
- D The disruption of peptide bonds of amino group and carboxyl group of amino acid residues.

4) Which statement about triglycerides is correct?

- A They are made up of three fatty acids combined with glycogen.
- B They are more saturated with hydrogen compared with phospholipids.
- C They form a bilayer in the membranes of cells.
- D They have a lower ratio of oxygen to carbon compared with carbohydrates.

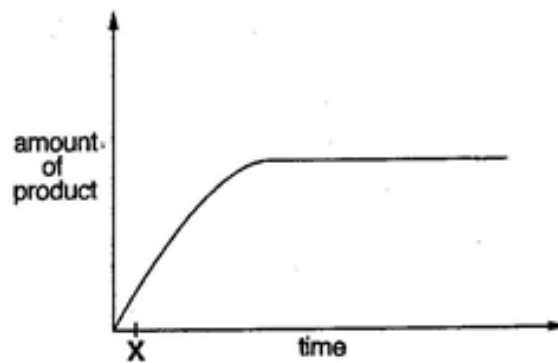
5) Diagrams 1, 2 and 3 show the structural formulae of three polysaccharides.



What are the names of these polysaccharides?

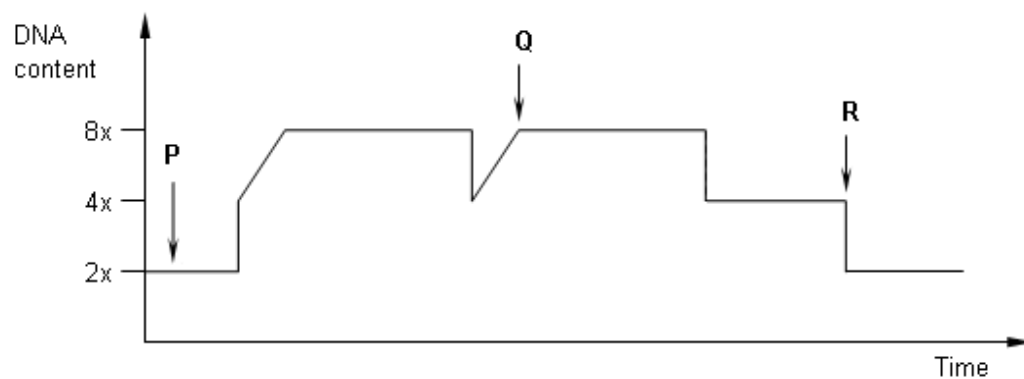
	1	2	3
A	Amylose	Cellulose	Glycogen
B	Amylose	Glycogen	Amylose
C	Cellulose	Glycogen	Cellulose
D	Glycogen	Amylose	Amylose

- 6) The graph below shows the amount of product formed in an enzyme-catalysed reaction over a certain period of time at 37° C.



What is true at time X?

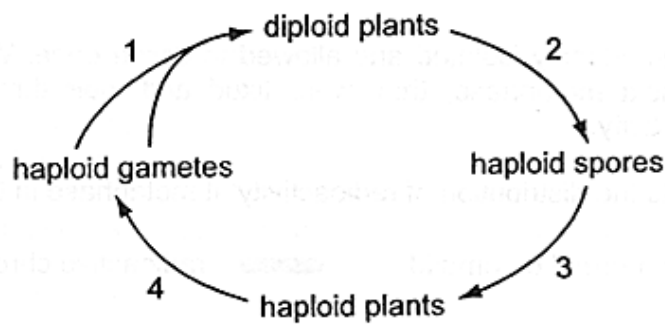
- A Most enzyme molecules will have free active sites.
 - B The number of unreacted substrate molecules is high.
 - C The number of enzyme-substrate complexes is low.
 - D The rate of enzymatic reaction is low.
- 7) The graph below represents the changes in the DNA content in a cell at different stages in the cell cycle.



At point P, the number of chromosomes in a cell is 20. What is the number of chromosomes in a cell at point Q and what are the events occurring at points Q and R?

	Number of chromosomes in the cell at point Q	Events at point Q	Events at point R
A	40	G ₂ phase	Cytokinesis
B	80	Prophase I	Telophase II
C	40	Prophase I	Telophase II
D	80	G ₂ phase	Cytokinesis

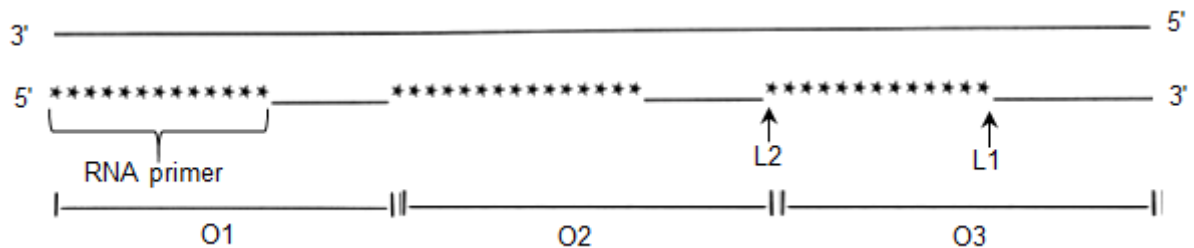
8) The diagram shows the life cycle of a type of plant.



When does meiosis occur?

- A 1 only
- B 2 only
- C 2 and 3 only
- D 2, 3 and 4 only

9) The figure below shows the formation of a lagging strand from its DNA template before the removal of the RNA primer.



Which row correctly shows the events taking place during the synthesis of the lagging strand?

	First Okazaki fragment to be synthesised	Site of catalysis by DNA ligase
A	O1	L1
B	O1	L2
C	O3	L1
D	O3	L2

- 10)** In a pre-mRNA molecule, 36% of the total nitrogenous bases present are adenine bases and 36% of the total nitrogenous bases present are uracil bases.

Which of the following best estimates the percentage of guanine in the template strand of the segment of DNA that was used for the synthesis of the pre-mRNA molecule?

- A** 14%
- B** 28%
- C** More than 28%
- D** Less than 28%

- 11)** The diagram below shows the entire coding region of an mRNA molecule with the codon AUG coding for the first amino acid.

5' – AUG CAC AAA GAC GGC CCG GCA CCC CCU AGA AAA UAA – 3'

A mutation occurred which resulted in a mutant polypeptide of 8 amino acids instead of the normal 11 amino acids. The sequence of the 8 amino acids in the mutant polypeptide and the sequence of the first 8 amino acids in the normal polypeptide were the same.

Assuming that the DNA contained no introns, which of the following mutation(s) could by itself result in the production of the mutant polypeptide?

- I.** Triple base insertion
- II.** Double base transition
- III.** Double base transversion
- IV.** Double base deletion

- A** I only
- B** I and III only
- C** I and IV only
- D** II, III and IV only

- 12)** Rhabdoviruses infect human cells. The genome of rhabdoviruses consists of a single-stranded RNA molecule whose sequence is complementary to the RNA sequence which functions as a messenger RNA.

How is the “+” messenger RNA produced in such cells by rhabdovirus?

- A** Host cell RNA polymerase activity.
- B** One portion of the infecting RNA is directly translated by host cell ribosomes.
- C** Reverse transcriptase activity.
- D** The infecting virus particle contains an RNA-dependent RNA polymerase.

13) In *E. coli*, the production of enzymes for tryptophan synthesis is carefully controlled according to the organism's metabolic needs. A mutation in the gene encoding the tryptophan repressor has occurred, such that the repressor can bind DNA without the co-repressor. What effect on enzyme production can be expected under this condition?

- A** High-level enzyme production under any condition.
- B** High-level enzyme production in the absence of tryptophan, no activity in the presence of tryptophan.
- C** No enzyme production in the absence of tryptophan, high-level activity in the presence of tryptophan.
- D** No enzyme production under any conditions.

14) A scientist is using an ampicillin-resistant strain of bacteria which has a non-functional lacZ gene.

She has a F plasmid containing a functional copy of lacZ gene. She then adds a high concentration of the plasmid to a tube of the bacteria in a medium for bacterial growth that contains glucose or lactose as the only energy source. This tube (+) and a control tube (-) with similar bacteria but no plasmid are both incubated under the appropriate conditions for growth and plasmid uptake. The scientist then spreads a sample of each bacterial culture (+ and -) on each of the three types of plates indicated below.

Plate A: Glucose medium

Plate B: Glucose medium with ampicillin

Plate C: Lactose medium with ampicillin

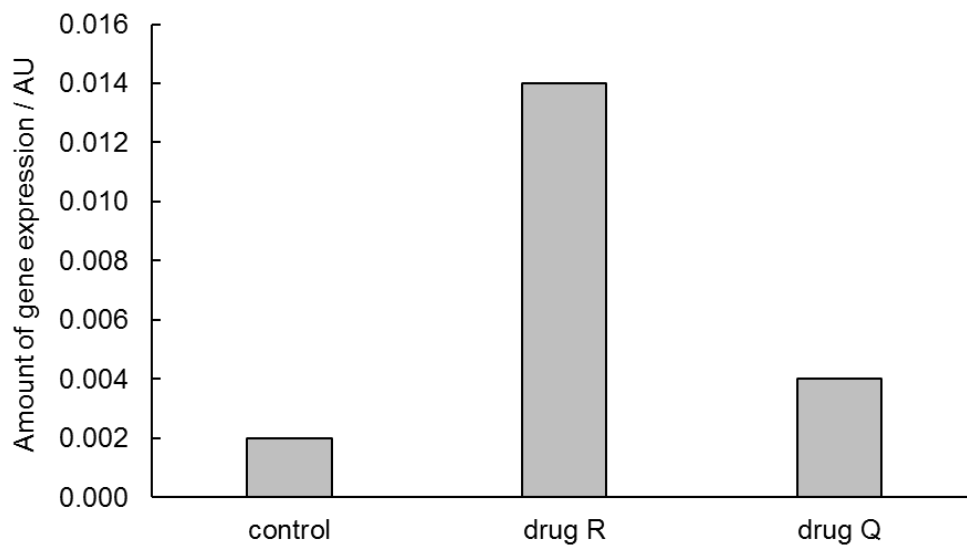
Which row correctly identifies the plates with growth (✓) or no growth (✗)?

	Bacterial strain with added plasmid (+)			Bacterial strain with no plasmid (-)		
	Plate A	Plate B	Plate C	Plate A	Plate B	Plate C
A	✓	✓	✓	✓	✓	✓
B	✓	✓	✓	✓	✓	✗
C	✓	✗	✗	✗	✗	✗
D	✗	✗	✓	✗	✗	✗

15) Which of the following is not a requirement for the mechanism of RNA splicing?

- A** Formation of a pre-mRNA-snRNP complex.
- B** Involvement of guanylyl transferase.
- C** 5' and 3' splice sites.
- D** Complementary base pairing between snRNAs with conserved sequences.

16) Drug R is a DNA methyltransferase inhibitor and drug Q is a histone deacetylase inhibitor. An experiment was carried out to investigate the effects of drug R and Q on expression of a gene. The graph shows the experimental results.



Which are possible explanations to the results shown?

- I.** Drug Q controls modification of histones, resulting in weaker binding of histones to DNA.
- II.** Drug Q increases gene expression by increasing accessibility of RNA polymerase to the promoter.
- III.** Drug R increases gene expression by preventing methylation of DNA.
- IV.** Inhibiting DNA methylation is more effective in increasing gene expression than inhibiting histone deacetylation.

- A** I and II only
- B** II and IV only
- C** I, II and III only
- D** All of the above

- 17)** Cells taken from a human bone cancer multiplied readily in culture. Analysis showed that the cells were unable to produce the protein, RB.

Addition of RB to these cells reduced their rate of division.

Which of the following statement accounts for the observation?

- A** Both chromosomes in the cancer cell carry alleles for tumour suppressor gene.
- B** Both chromosomes in the cancer cell have the allele for tumour suppressor gene deleted.
- C** Both chromosomes in the cancer cell carry alleles for proto-oncogene.
- D** Both chromosomes in the cancer cell have the allele for proto-oncogene deleted.

- 18)** A fisherman was surprised to catch a fish which had no scales (nude). To investigate the origin of this phenotype the nude fish was mated several times to fish with scales and the result of each cross was recorded. In the crosses of nude with scaled, a third phenotype appeared, which was later called linear. The linear phenotype has only a single line of scales down one side of the body.

The outcomes of these crosses are shown in the table.

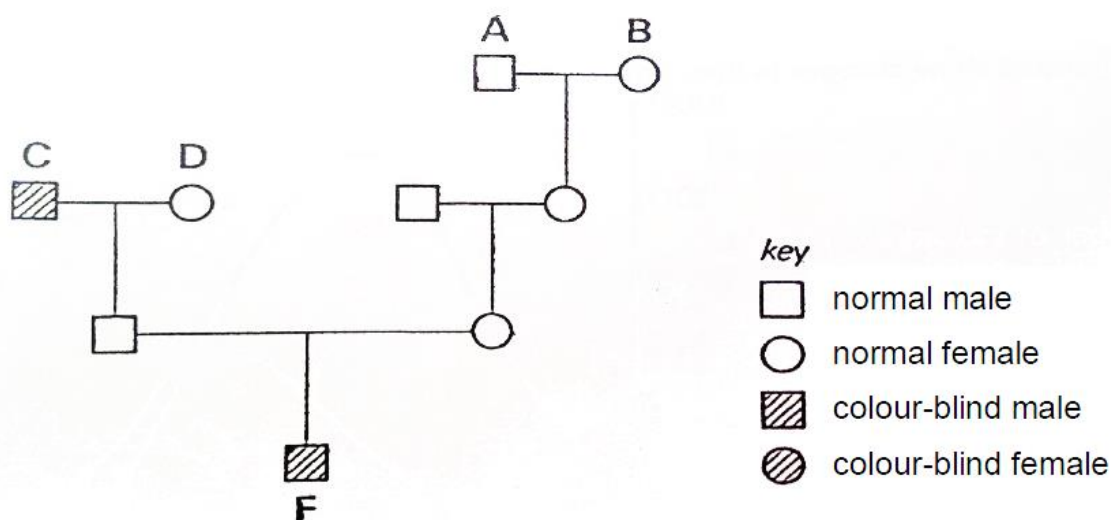
Cross	Parents	Offspring phenotype and ratio
1	scaled x nude	all offspring linear
2	linear x linear	1 scaled : 2 linear : 1 nude

Which of the following is conclusive from the above data?

- A** There is incomplete dominance between the nude and scaled phenotype.
- B** The environment is the reason for the loss of scales in the nude fish.
- C** All of the linear fish are homozygous.
- D** The nude fish are heterozygous.

- 19) The diagram below shows the pedigree of a family carrying sex-linked allele for colour-blindness.

From which labeled member of his family did F inherit this disorder?



- 20) In *Primula*, the dominant allele, A, of a gene for anthocyanin pigment production gives magenta-coloured flowers. The recessive allele, a, gives yellow-coloured flowers. The dominant allele, B, of another unlinked gene inhibits the effect of allele A; while the recessive allele, b, has no effect. Plants in which allele A is inhibited by allele B have yellow flowers.

A yellow-flowered plant is known to be homozygous recessive at the A/a locus. The genotype at the B/b locus is unknown.

Which of the following genotype of the test cross will reveal the genotype of the yellow-flowered plant?

- A aabb
- B AABB
- C AAbb
- D aaBB

- 21)** A blue bird with red eyes with the genotype BBrr was crossed with a white bird with black eyes with the genotype bbRR. The F₁ birds were allowed to cross-fertilised. A chi-squared test was carried out on the results obtained for the F₂ generation which showed the phenotypic ratio of 12:3. Part of the table of values for χ^2 is shown.

degree of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

The value χ^2 in this investigation was 6.36.

Which of the following conclusion made about the inheritance of bird plumage colour and eye colour is correct?

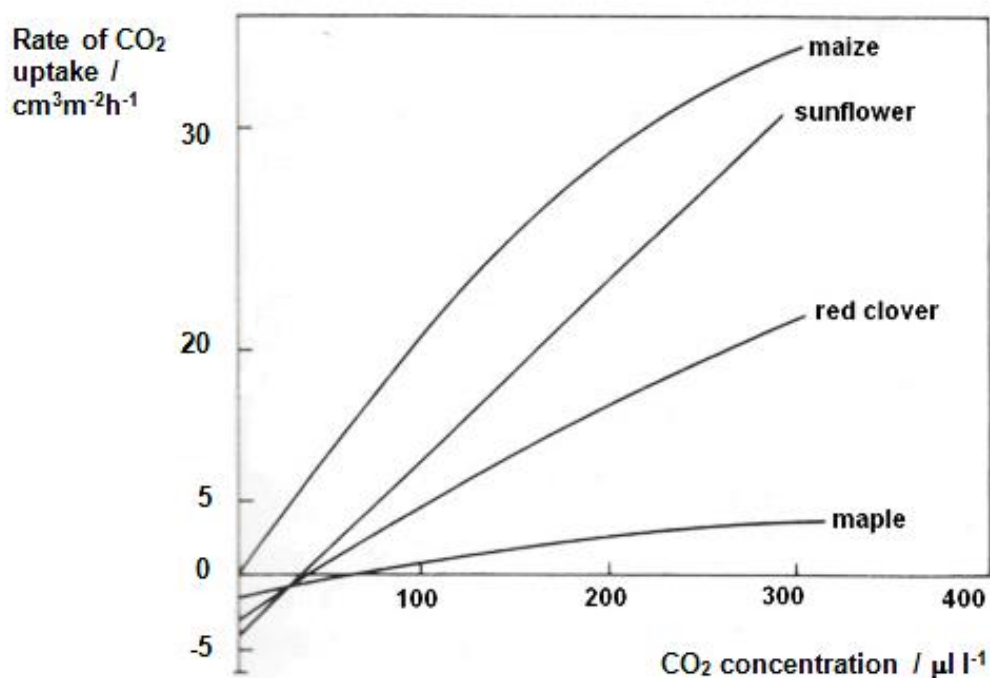
	probability, p	difference between observed and expected results
A	between 0.01 and 0.05	not significant
B	between 0.01 and 0.05	significant
C	between 0.02 and 0.05	not significant
D	between 0.02 and 0.05	significant

- 22)** Which of the following statements are true with regards to cyclic and non-cyclic phosphorylation?

- I. Hydrolysis of ATP occurs in both cyclic and non-cyclic phosphorylation.
- II. Energy released from the electron transport chain is used to pump protons from the stroma into the thylakoid lumen.
- III. NADP⁺ is oxidized in non-cyclic phosphorylation.
- IV. The products of non-cyclic phosphorylation are NADPH, ATP and oxygen.

- A** I and IV only
- B** II and III only
- C** II and IV only
- D** II, III and IV only

- 23) The graph below shows the rate of photosynthesis of four different plants at different concentrations of carbon dioxide.



Which of the following conclusions can be made?

- I. At CO_2 concentrations below $150 \mu\text{l l}^{-1}$, CO_2 concentration is the main limiting factor for all the plants.
- II. CO_2 compensation point is around $40 \mu\text{l l}^{-1}$ for sunflower and red clover, and it measures the light intensity when the rate of CO_2 uptake equals to the rate of CO_2 given off.
- III. Maize has almost zero rate of respiration.
- IV. Maple has the lowest amount of organic compound produced at CO_2 concentration of $200 \mu\text{l l}^{-1}$.

- A I and IV only
- B I, III and IV only
- C II, III and IV only
- D II and III only

24) Mammalian liver cells were homogenised and the resulting homogenate centrifuged. Portions containing only mitochondria and cytosol (residual cytoplasm) were each isolated. Samples of each portion, and of the complete homogenate, were incubated in four ways:

- 1 with glucose;
- 2 with pyruvate;
- 3 with glucose plus cyanide; and
- 4 with pyruvate plus cyanide.

Cyanide inhibits carriers in the electron transport chain. After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in the table below.

	samples of homogenate					
	complete		mitochondria only		cytosol	
	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate
1 glucose	✓	✓	×	×	×	✓
2 pyruvate	✓	✓	✓	×	×	✓
3 glucose + cyanide	×	✓	×	×	×	✓
4 pyruvate + cyanide	×	✓	×	×	×	✓

× = absent, ✓ = present

Which statement can be concluded from the table?

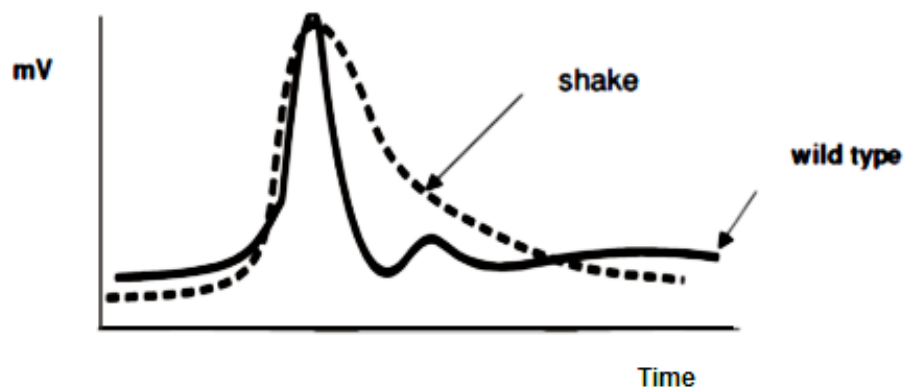
- A** Carbon dioxide was not formed when mitochondria was incubated with glucose as there was no oxygen present.
- B** Both aerobic and anaerobic respirations were occurring in **3**.
- C** Lactate formation in mitochondria was inhibited by the presence of cyanide.
- D** The mode of action of cyanide would be similar to that of a cell experiencing anaerobic respiration.

- 25) Trematol is a metabolic poison derived from the white snake root. Cows eating this plant concentrate the poison in their milk. The poison inhibits liver enzymes that convert lactate to other compounds for metabolism.

Which row illustrates the events that occur in an exercising athlete who consumed the trematol-tainted milk?

	lactate accumulation	NAD production	ATP production	pH of blood
A	yes	yes	yes	decreased
B	no	yes	yes	decreased
C	yes	no	no	increased
D	no	no	no	increased

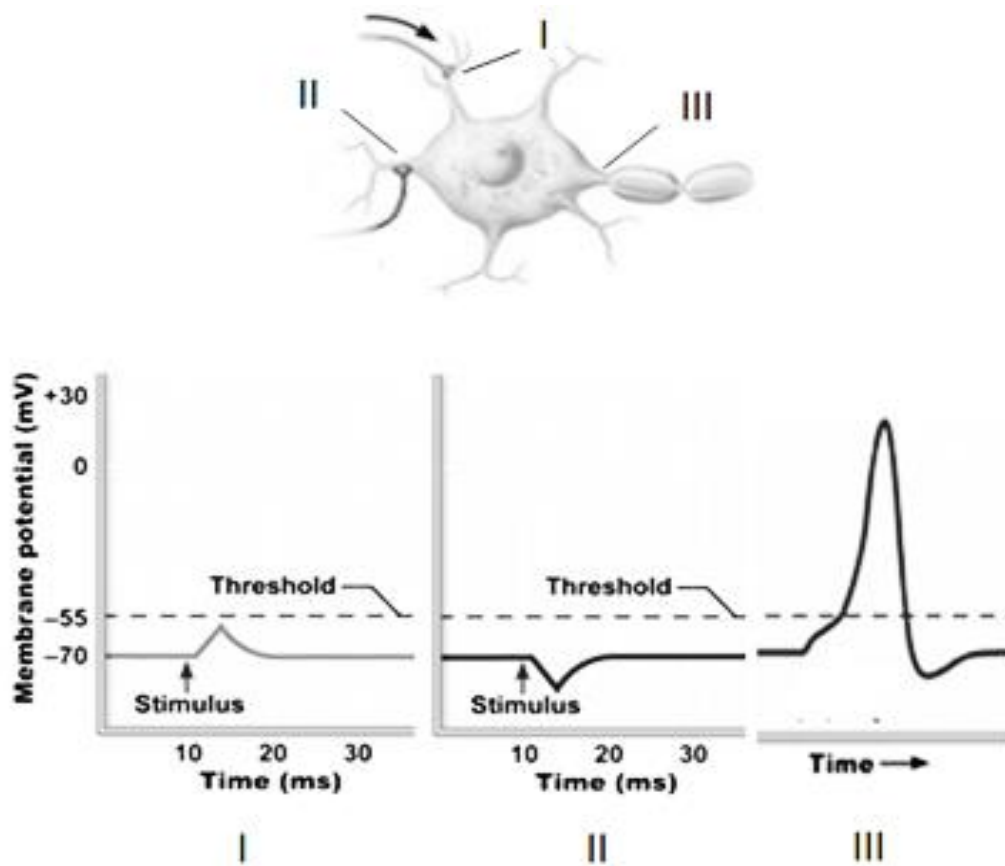
- 26) *Drosophila* flies homozygous for the *shake* mutation are extremely sensitive to diethyl ether that causes convulsions in homozygous individuals. Convulsions are caused by abnormalities in nerve impulse conduction as shown in the graph below.



Which of the following structures is impaired in the *shake* mutations?

- A Na^+ channels
- B K^+ channels
- C Ca^{2+} channels
- D Na^+/K^+ pump

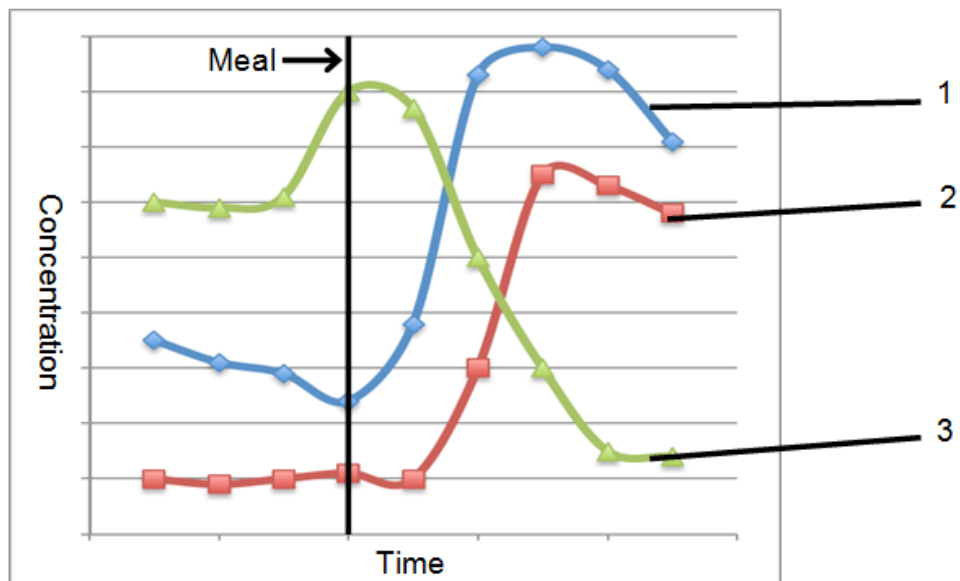
- 27) A neuron receives inputs from 2 other neurons as illustrated in the diagram below. The graphs show the membrane potential at points I, II and III on the post-synaptic neuron.



If the voltage-gated calcium channels on the pre-synaptic membrane at point II are no longer able to respond to a depolarisation, what is the expected effect on the post-synaptic neuron?

- A The post-synaptic neuron is now unable to receive any inputs.
- B Post-synaptic neuron is less depolarised and a stronger than normal stimulus is required to initiate an action potential in the post-synaptic neuron.
- C Post-synaptic neuron is more depolarised and a weaker than normal stimulus is sufficient to initiate an action potential in the post-synaptic neuron.
- D Post-synaptic neuron is hyperpolarised and a weaker than normal stimulus is sufficient to initiate an action potential in the post-synaptic neuron.

- 28) The graphs below show the changes in the concentrations of glucose, insulin and glucagon in the blood after a meal.



Identify what graphs 1-3 correspond to.

	1	2	3
A	Glucose	Insulin	Glucagon
B	Insulin	Glucose	Glucagon
C	Glucagon	Insulin	Glucose
D	Glucose	Glucagon	Insulin

- 29)** A scientist is investigating the effects of Poison T on the cell signaling pathway of glucagon. It is found that Poison T diminishes the effect of glucagon.

The table below shows the different components of the cell signaling pathway of glucagon and their statuses.

Component	Status
G Protein Coupled Receptor (GPCR)	Configuration can be changed
G protein	Can exchange GDP for GTP
cAMP levels	low
Protein Kinase A	Inactivated

Which of the following statements can be concluded from the results?

- I. Poison T interferes with reception.
- II. Poison T prevents G protein from hydrolysing GTP.
- III. Poison T inactivates the enzyme adenylate cyclase.
- IV. Poison T stops the relaying of message down the phosphorylation.

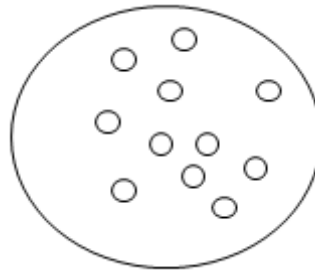
- A I and II
- B II and III
- C II and IV
- D III and IV

- 30)** Which of the 2 primers can be used to amplify the sequence shown below by Polymerase Chain Reaction?

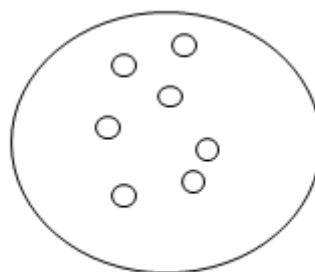
5' – GGAATTCGT -- // -- TATGCTACC – 3'

- A 5' GGTAGCATA 3' & 5' GGAATTCGT 3'
- B 5' ACGAATTCC 3' & 5' TATGCTACC 3'
- C 5' TATGCTACC 3' & 5' GGAATTCGT 3'
- D 5' ACGAATTCC 3' & 5' GGTAGCATA 3'

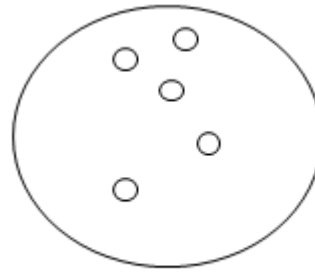
- 31)** The recombinant plasmids were transformed into *Escherichia coli* and the bacteria were grown on nutrient medium. The resulting master plate is shown in the diagram.



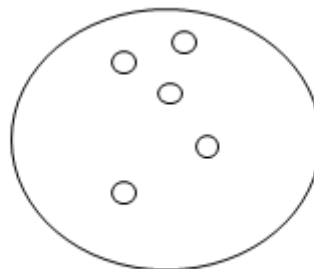
Transformed cells were selected by replica plating and grown on media containing different antibiotics.



medium with ampicillin



medium with tetracycline

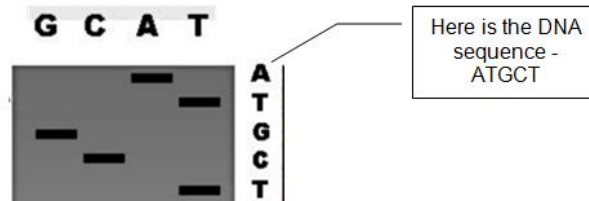


medium with ampicillin and tetracycline

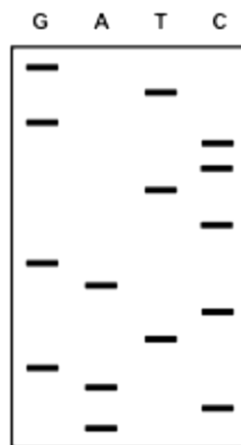
How many colonies in the master plate contain the recombinant plasmid?

- A** 1
- B** 2
- C** 5
- D** 7

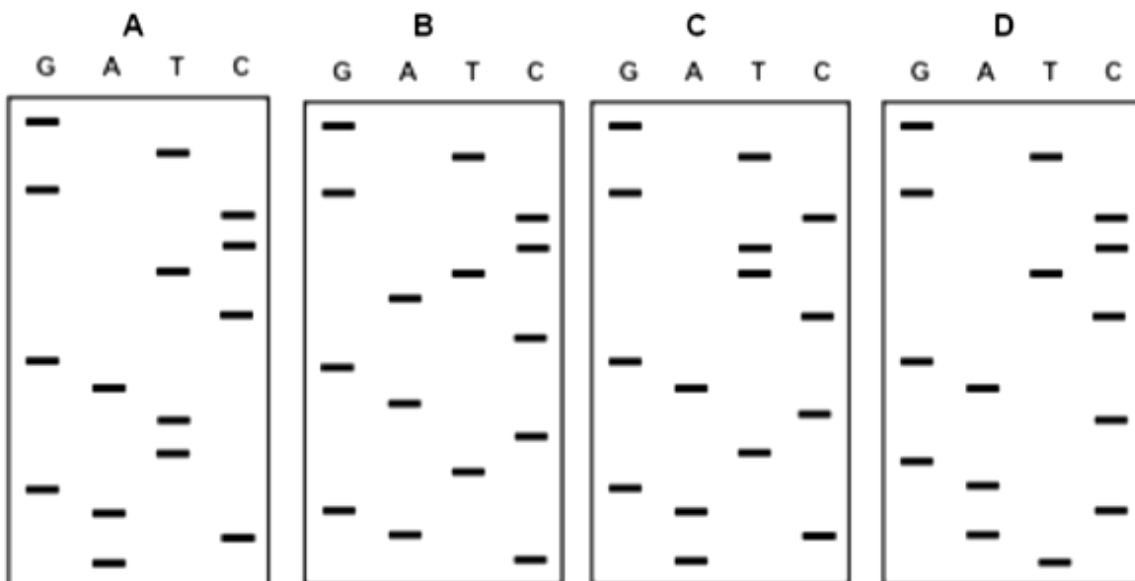
- 32) In people with a gene mutation where a base insertion has occurred, the protein formed has a very different primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using gel electrophoresis. Read from the top to the bottom for the DNA sequence. Here's an example:



The electrophoresis result from the DNA of a normal allele of the gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?



33) Genes P, Q, R and S occur on the same chromosome.

The table shows the recombination frequencies.

	recombination frequency (%)
between P and Q	46
between P and R	8
between R and Q	52
between Q and S	13
between R and S	41

Which of the following represents the correct order of genes on the chromosome?

- A** PQRS
- B** PRSQ
- C** RPSQ
- D** RSQP

34) Totipotency is demonstrated when _____.

- A** mutations in genes result in the development of misplaced appendages.
- B** a cell isolated from a plant leaf grows into a normal adult plant.
- C** an embryonic cell divides and differentiates.
- D** the replacement of the nucleus of an unfertilized egg with that of an intestinal cell converts the egg to an intestinal cell.

35) One therapy to treat β -thalassaemia is to transplant bone marrow cells from a genetically compatible donor into a patient. A potential gene therapy involves adding the normal, dominant allele for β -globin to the patient's cells.

Which of the following would ensure that the normal gene is passed on to the next generation?

- A** Using a retrovirus to introduce the normal β -globin gene into bone marrow cells.
- B** Using a retrovirus to introduce the normal β -globin gene into an egg cell.
- C** Using an adenoviral vector to introduce the normal β -globin gene into bone marrow cells.
- D** Using an adenoviral vector to introduce the normal β -globin gene into an egg cell.

36) Corn is a major crop grown in Europe. In the past, it was either ruined by attack from the corn borer larvae or intensively sprayed with pesticides each year. A biotech company gained approval to insert a Bt gene into corn. This gene codes for a protein that kills the larvae feeding on the corn. The genetically engineered Bt corn initially thrived without the addition of any pesticides, but later suffered damage from pests again.

Which of the following are possible reasons why the Bt corn is again under attack from pests?

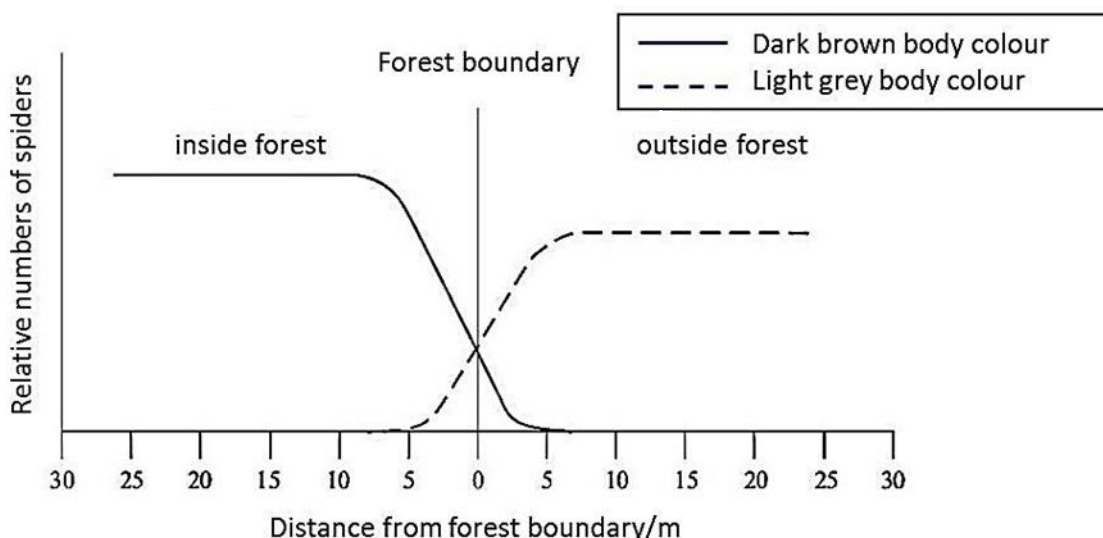
- I. A strain of resistant larvae has emerged as a chance mutation. The frequency of the gene conferring resistance as well as the number of larvae with resistance has increased. The Bt gene is now ineffective.
- II. Corn is being attacked by other pests which are not killed by the Bt gene protein.
- III. The Bt allele has undergone insertional mutagenesis. This mutagenesis has had a selective advantage in seed fertility.

- A I and II only
- B I and III only
- C II and III only
- D All of the above

37) Which taxonomic group is correctly described?

	Taxonomic group	Size of group	Number of similarities between members of the group	Ancestry of members of group
A	kingdom	large	few	recently shared a common ancestor
B	species	small	many	recently shared a common ancestor
C	species	large	few	distantly shared a common ancestor
D	kingdom	small	many	distantly shared a common ancestor

38) The graph below shows the distribution of huntsman spiders at a forest boundary.



One species of huntsman spider (*Isopeda isopedella*) varies in body colour from dark brown to light grey. In one community at the forest boundary, two populations of this species were found. Some were found living in the leaf litter inside the forest and others were found living in the grass just outside the forest. The relative numbers of dark brown adult spiders and light grey adult spiders found at certain distances from the forest boundary are shown in the graph above.

Which of the following would be the best explanation for this distribution?

- A The two populations of spiders were once different species.
- B The two populations of spiders were unable to interbreed and individuals were adapting to suit their habitats.
- C The differences in the two habitats had changed the physical appearance of individual spiders.
- D A particular body colour provided a selective advantage to spiders in a particular habitat.

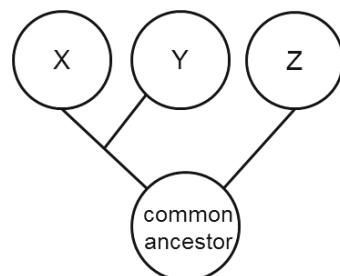
39) Which of the following statements does not correctly compare the neutral theory of molecular evolution and natural selection?

- A Neutral theory of molecular evolution accounts for most of the differences that we observe at the phenotypic level as compared to natural selection.
- B Neutral theory of molecular evolution accounts for most of the differences that we observe at the genotypic level as compared to natural selection.
- C The rate of change in the nucleotide sequence brought about by neutral theory of molecular evolution occurs at a constant rate due to random chance events while the rate of change brought about by natural selection can be fast or slow depending on the strength of the selection pressure.
- D Neutral theory of molecular evolution is largely responsible for the RFLP that we observe between species as compared to natural selection.

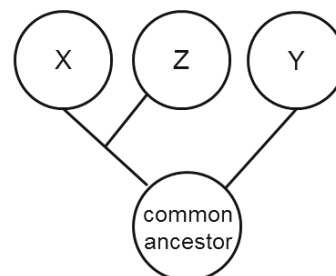
- 40) Cytochrome c is a protein found in most organisms. The amino acid sequence of this protein varies between different species and can be used to determine evolutionary relationships. The table shows the numbers of differences in the amino acid sequences of cytochrome c between three species (X, Y, and Z).

	species Y	species Z
Species X	8	2
Species Y	0	9

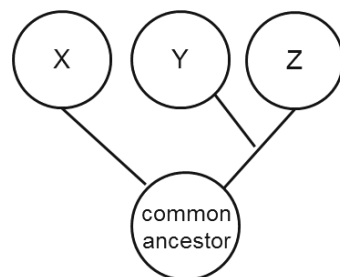
Which diagram best represents the evolutionary relationships between species X, Y and Z?



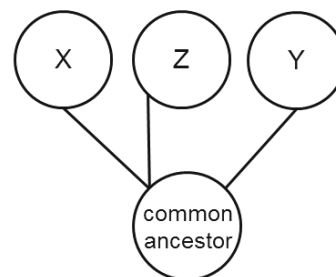
A



B



C



D

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